

Two Core Projects

D1: experimental design | D2: evaluation without leakage

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Why these projects?

These two projects ask you to make and defend decisions rather than assemble an impressive but opaque pipeline. Every required computation is provided in a small notebook. Your intellectual work is to choose the appropriate experimental unit, inspect assumptions, compare alternatives and write a defensible conclusion. Each project contributes 12% of the course grade.

Feature	D1	D2
Launch / deadline	Week 5 / end Week 6	Week 9 / end Week 10
Biological question	How should we study a treatment response?	Will a classifier generalize to new donors?
Core data	Hypothetical costs/yields and synthetic examples	Synthetic repeated donor profiles and random labels
Deliverables	Notebook + design memo + audit	Notebook + evaluation memo + audit
Length / workload	<=1200 words, excluding tables/references; about 6 hours	<=1200 words, excluding tables/references; about 6 hours

AI and peer discussion are permitted for explanation/debugging, with disclosure. Individual reasoning and checks are required. No patient data, paid API, advanced model training or original software package is expected. Correctly reporting an inconclusive result can earn full marks.

D1 | Design a treatment-response RNA-seq experiment

Scenario: You want to compare RNA abundance in a fictional cultured-cell system with and without a treatment. You can prepare independent cultures under each condition. There is no pilot variance estimate yet. You have a hypothetical budget of 6000 units, two processing batches and a provider offering the fictional costs below. This is an educational design problem, not a quotation or a validated power calculation.

Assumption	Value
Library preparation	150 budget units per sample
Sequencing	10 units per million paired-end fragments
Usable fraction	0.80 of purchased read pairs; sensitivity scenario0.60
Experimental groups	Two conditions
Budget	6000 units

Assumption	Value
Candidate designs	A:2 cultures/group,80 million pairs/sample; B:4/group,30 million; C:6/group,15 million

Required decisions

- State a biological question and a measurable RNA-level outcome; explain the boundary to protein/function claims.
- Compare A, B and C by total cost, independent sample count and usable pairs per sample. Consider the 0.60 usable-fraction scenario as a robustness check.
- Recommend a design or a justified alternative. State what pilot information is missing before a formal power analysis. There is no uniquely proven optimum.
- Allocate both conditions across both batches, with independent cultures and randomized processing order. Identify a positive/negative or procedural control appropriate to the question without pretending it replaces biological replication.
- Propose prospective QC/exclusion criteria, record missing samples, preserve metadata and outline a supported count-based differential-expression analysis with multiplicity control. Do not perform fake DE on the two-gene excerpt.

Do not confuse these two calculations

The notebook also computes nominal depth for a separate toy DNA genome: one million read pairs x two ends x 150 bases / 5 million target bases = 60x. Multiplying by 0.80 gives 48x usable mean-depth estimate. This checks units only. It is not a sufficient read-depth prescription for a transcriptome, a minimum per-base coverage or an RNA-seq power analysis. [S03,S09,S10]

Memo structure and required artifacts

Section	Expected evidence
Question and measurement (~150 words)	System, treatment/control, unit and outcome.
Design and allocation (~300 words)	Comparison table; chosen design; two-batch allocation table; randomization/controls.
Calculations and uncertainty (~250 words)	Costs, usable yields at 0.80 and 0.60; assumptions; sensitivity analysis.
Analysis and interpretation (~250 words)	Metadata match, count model/design, effect sizes, uncertainty, BH family, limits.
Audit and reflection (~150 words)	GENOMES record, one check/correction or confirmed check, sources, help/AI disclosure.

Submit: W05 notebook with outputs and responses; exported summary JSON; memo as PDF or DOCX; the audit can be included in the memo. Filename: D1_course-pseudonym. The draft

checkpoint in Week 5 is a one-paragraph question plus a design table; feedback is formative. Submit the finished project at the end of Week 6.

D1 analytic rubric

Criterion	Points	Full-credit evidence
Question and independent unit	15	A measurable RNA-level question, correct independent cultures and distinction from technical repeats.
Design, controls and allocation	25	Balanced conditions across batches; credible randomization/control plan; prospective QC; a reasoned choice.
Cost and read-yield calculations	20	Correct units and arithmetic for all options; 0.80/0.60 sensitivity; no RNA-power claim from genome depth.
Analysis and limitations	20	Supported count-based inference, sample matching, multiplicity, missing pilot information and bounded conclusions.
Reproducibility and sources	10	Clean rerun or documented paper route; GENOMES; provenance and appropriate source support.
Communication and ownership	10	Clear memo, own explanation and honest assistance disclosure; no penalty for using no AI.

D2 | Is the model learning disease or recognizing donors?

Scenario: A fictional developer claims that a model can identify a condition from repeated molecular profiles. The dataset contains 12 donors with 6 rows each and 8 features. Each donor has a distinctive synthetic signature. Donor labels are randomly assigned, with no designed population-level disease signal. A simple nearest-neighbour classifier is provided. Your job is to audit evaluation, not invent a better neural network.

Required investigation

- Compare a within-donor split (four rows per donor for training, two for test) with a donor-disjoint split (eight donors for training, four for test). Draw the split and count the independent test units.
- Explain where standardization is fitted and why test information must not influence it. The provided model fits the scale only on training rows.
- Report row-level and donor-level accuracy. Explain the predeclared donor majority-vote tie rule. Discuss class balance and the limitations of only four test donors.
- Summarize the 100 conditional donor-level label permutations. Labels are shuffled within the predeclared training and test partitions, preserving class counts; all rows of a donor keep one label. This is a conditional null reference, not cell-wise randomization or clinical validation.
- Recommend an external evaluation: independent donors, a suitable simple baseline, grouped model-selection procedure, untouched final test and relevant site/time differences. Report uncertainty and subgroup failures.

What is not a correct objective

Do not search pseudonyms/seeds until the result tells a preferred story. Keep the assigned seed. Random labels can yield chance finite-sample associations; a donor holdout can occasionally perform well. Full marks depend on correct design, reported outputs and reasoning, not a mandatory accuracy drop or a value of exactly 0.50. [S11]

Memo structure and required artifacts

Section	Expected evidence
Claim and population (~150 words)	State whether the target is new rows from known donors or entirely new donors.
Data and split (~300 words)	Data provenance; donor/row counts; split diagram; training-only preprocessing.
Results (~250 words)	Row and donor metrics for each split, permutation reference and class balance.
Interpretation (~250 words)	Leakage, uncertainty, random-label simulation truth and a proposed external evaluation.

Section	Expected evidence
Audit and reflection (~150 words)	GENOMES, one check, sources, AI/no-AI disclosure and limitations.

Submit: W09 notebook, summary JSON, memo as PDF or DOCX with a split diagram/table.

Filename: D2_course-pseudonym. Week 9 checkpoint: propose the split before inspecting results.

Final submission is due end of Week 10. A hand-drawn split is acceptable; artistic quality is not assessed.

D2 analytic rubric

Criterion	Points	Full-credit evidence
Question and evaluation unit	15	Clear future population and correct independent donor count.
Split and preprocessing	25	Donor-disjoint evaluation for new-donor claims; transformations fitted only on train; no test-driven tuning.
Metrics and result reporting	20	Accurate row/donor outputs, class balance and predeclared aggregation; no omitted inconvenient results.
Null reference and uncertainty	20	Correct donor-level conditional permutation explanation; four-donor limitations; no forced benchmark conclusion.
Reproducibility and sources	10	Seed, versions, data status, clean rerun/auditable paper route and source support.
Communication and ownership	10	Clear split and bounded conclusion, independent explanation, honest assistance disclosure.

How partial credit is applied

For each criterion, full credit requires the described evidence. A mostly correct response with a minor omission typically earns about 75%; a partial response with a material gap about 50%; absent or fundamentally unsupported evidence earns 0-25%. Markers can use intermediate points. A single arithmetic slip should not erase correct biological reasoning, and the same underlying defect should not be repeatedly penalized across unrelated criteria. Contradictory claims must be resolved, not hidden by polished prose.

The individual oral explanation

The oral uses your D1 and one GENOMES check. You will be asked to identify the independent unit, explain a number/decision and state a limitation or verification step. It is not a demand to memorize code. You may refer to your own submitted project. Three prompts are scored 0-3 each and converted to $100 \times \text{score}/9$. The final course contribution is 18% of that percentage. The instructor schedules individual slots and approved accommodations.

Optional portfolio, not a third graded project

At the end, collect D1, D2, one corrected weekly task and a one-page reflection: What claim would you now refuse to make, and what evidence would change your mind? This consolidates existing work without adding a compulsory research project. Never present synthetic outputs as experimental discoveries.